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Clinica Chimica Acta 372 (2006) 54–64
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Analytical performance of the Iris iQ200 automated urine microscopy analyzer

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Received 24 January 2006; received in revised form 13 March 2006; accepted 13 March 2006
Available online 11 May 2006

Abstract

Background: We evaluated the Iris iQ200 Automated Urine Microscopy Analyzer to find out if the instrument performed better than traditional visual bright field microscopy in detecting basic urine particles, as assessed against reference phase contrast microscopy.
Methods: The HUSLAB quality system was followed in planning and completing the evaluation process. The iQ200 instrument results from 167 mid-stream, uncentrifuged urine specimens were compared to those obtained with phase contrast reference microscopy, and to those with routine bright field microscopy. Linearity, carry-over and precision were tested according to well-established protocols.
Results: The iQ200 counted erythrocytes (RBC) at $r=0.894$ ($R^2=0.799$) with Automated Particle Recognition (APR) software alone and at $r=0.948$ ($R^2=0.898$) after re-classification. The performance for leukocytes (WBC) was $r=0.885$ with APR and $r=0.978$ after re-classification. The correlations of counting after user re-classification were $r=0.927$ for squamous epithelial cells (SQEP), $r=0.856$ for casts, and $r=0.706$ for non-squamous epithelial cells. The iQ200 showed good linearity and precision and no carry-over was detected.
Conclusions: The Iris iQ200 was capable to count reliably RBC, WBC, and SQEP cells and to identify a fraction of bacteria and renal elements. Counting results equalled or exceeded that of routine bright field microscopy or earlier flow cytometric technology. The instrument eliminates manual sample preparation but requires a well-trained technologist for re-grouping of findings.
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Keywords: Automation; Urine analysis; Urine particles; Urine microscopy

1. Introduction

The *Iris iQ200 Urine Microscopy Analyzer* is a second-generation automated urine analyzer developed by Iris Diagnostics Inc. (Chatsworth, California, USA), the pioneer of automated urine microscopic instruments since the 1980's [1]. Since then, flow cytometry technology was introduced to automated urine particle analysis [2,3]. As are previous Iris instruments, the Iris iQ200 is based on capturing images from planar flow of urine particles [4] with a CCD (charged coupling device) camera. The iQ200 software allows a trained technologist to display these images of specimen particles for visual confirmation. The iQ200 technology improves its walk-away automation. The iQ200 refined neural network algorithm APR (Automated Particle Recognition system) uses particle size, shape, contrast and texture as features to enhance its classification of detected particles. We assessed the performance of the iQ200 Urine Particle Analyzer against sophisticated visual

Abbreviations: APR, Automated Particle Recognition; ART, artefacts; BACT, bacteria; BYST, yeast; CCD, charged coupling device; CI, confidence interval; HYAL, hyaline casts; MUCS, mucus; NSE, non-squamous epithelial cells; RBC, red blood cells; SPRM, sperm; SQEP, squamous epithelial cells; UNCX, crystals; UNCC, non-hyaline casts; WBC, white blood cells; WBCC, white blood cell clumps.

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doi:10.1016/j.cca.2006.03.015

chamber counting of uncentrifuged fresh urine specimens obtained from patients representing several clinical specialties.

2. Materials and methods

2.1. Specimen collection and storage

The evaluation protocol followed the general principles of method validation defined in the operational laboratory procedures of HUSLAB quality system.

The mid-stream urine specimens were collected as a part of the medical examination from 167 out- or in-patients attending Helsinki University Central Hospital at the departments of paediatrics, oncology, surgery and internal medicine, and stored at +4 °C before analysis, during a period of two weeks. Specimens were selected based on findings of erythrocytes, leukocytes or epithelial cells in routine bright field visual microscopic analysis (see below for detail). This also provided an average performance of laboratory personnel at a Finnish university hospital for this evaluation. The actual comparison was carried out by using advanced *phase contrast* microscopy against automated Iris iQ200 analysis on the same day (see below). To balance the delays, the samples were first analyzed by phase contrast microscopy and then by iQ200 in the first week, while during the second week the order was reversed. The delay between the screening routine bright field microscopy and other investigations (i.e., iQ200 and phase contrast microscopy) was 1–3 h.

2.2. Iris iQ200 instrument, materials and technical detail

The following solutions and materials were provided by Iris Diagnostics Inc. for the evaluation: Iris iQ Calibrator (REF 475-0059), iQ Positive Control (REF 475-0046), iQ Negative Control (REF 475-0058), iQ Focus (REF 475-0060), Iris System Cleanser (REF 475-0003), Iris Diluent (REF 475-0021) and iQ Lamina (REF 475-0047). Calibrator, Focus and Positive Control are suspensions of fixed human red cells in a particle-free buffer. Negative Control is a particle free buffer solution. The Positive Control and Negative Control solutions were used to perform iQ200 system quality control. The iQ200 software includes quality control charts for negative and positive control suspensions that allow technologists to control counting accuracy over time. The iQ Lamina used to create the laminar flow is a particulate free isotonic buffer containing proprietary laminar flow stabilizers, bacteriostatic agents, fungicidal agents and preservatives.

The Iris iQ200 Automated Urinalysis System, software version 1.0 is composed of the iQ200 Automated Urine Microscopy Analyzer (iQ200), connected physically and electronically to the AUTION MAX AX- Automated Urine Chemistry Analyzer (*Arkray, Kobe, Japan*) based on reflectometry, and a computer workstation for reporting specimen composition results. The needed minimum urine specimen volume is 3 ml, 1 ml urine is aspirated into the instrument, and two microlitres are used for analysis. A urine specimen is hydro-dynamically focused between two layers of suspending fluid (lamina). Lamination is the planar equivalent of axial hydrodynamic focusing, traditionally used in many blood cell counters and in flow cytometers. Laminar flow allows particles to be seen with their largest projection facing the direction of lens. The specimen particles can be seen through the objective lens of a microscope coupled to a CCD video camera.

The CCD digital camera captures five hundred frames (1018 × 761 μm) per each sample; the volume is calculated from laminar fluid. Individual particle images are isolated within each frame. Size, shape, contrast and texture features are used by the APR based on neural network technology. After auto-classification, a trained operator may visually confirm the identification or re-classify the particles based on their morphological detail as displayed on the screen. The manufacturer recommends that a trained operator reviews the images of all abnormal results, and edits them if necessary. To speed up the editing process, the iQ200 software assorts particle images by auto-classified category. The categories reported by the instrument include red blood cells (RBC), white blood cells (WBC), white blood cell clumps (WBCC), squamous epithelial cells (SQEP), non-squamous epithelial cells (NSE), hyaline casts (HYAL), non-hyaline casts (UNCC), bacteria (BACT), crystals (UNCX), yeast (BYST), sperm (SPRM) and mucus (MUCS). Individual images that cannot be recognized with high confidence by the APR software are reported as artefacts (ART; in the later software called as unclassified particles). Particle counts are measured as particles per unit of volume based on the number of images and the volume scanned. Daily maintenance was performed as recommended by the manufacturer.

During the evaluation, original specimens were run with iQ200 (by P.R.) at HUSLAB. Classification was performed in the linearity, carry-over, and precision studies by applying APR only. In the procedure comparison study against visual phase contrast microscopy, digitally stored images were re-classified on screen as needed at Iris Corporation (by E.C.) to show the maximum obtainable performance. Image re-classification was performed without knowledge of test strip or phase contrast microscopy results.

2.3. Visual microscopy

As a reference for instrumental performance and routine microscopy procedure, sophisticated *phase contrast* microscopy was performed by one of the national reference laboratory technologists of urine particles (E.T.), working at HUSLAB. She used phase contrast optics at ×400 magnification operating with Nikon Eclipse E600 microscope. One-μl volumes of unstained and uncentrifuged specimens were counted in a Bürker chamber. Phase contrast optics was utilized to ensure detection of all RBC, bacteria and hyaline casts in the specimens. The recommended reference procedure [5] was made easier for faster process, by allowing the technologist to count a maximum of 50 epithelial cells or casts or a maximum of 200 RBC or WBC, or the total count of particles seen in the 1 μl volume even if any maximum number was not reached. This statistical imprecision was thus allowed in counts for practical reasons. Equivocal classifications were confirmed by staining the specimens with Sternheimer

supravital stain [6]; this was used in particular for correctly identifying casts or epithelial cells in the specimens. The staining was performed after centrifugation of a 10 ml sample at 400 ×*g* for 5 min, after which 0.5 ml resuspended sediment was stained with 50 µl of Sternheimer stain for 5 min. Finally, an aliquot of 13 µl was investigated under an 18 × 18 mm coverslip at ×400 magnification. The routine bright field visual microscopy was performed by several laboratory technologists at HUSLAB. Opposite to the reference procedure above, one-µl uncentrifuged urine was counted in a Bürker haemocytometer chamber by using bright field optics only.

2.4. Linearity, carry-over, and precision

The linearity of the instrument was tested by diluting iQ Positive Control solution (fixed RBC), and patient samples with high RBC or WBC counts with the iQ Diluent solution. The linearity tests were run as duplicates.

Carry-over test was performed according to a protocol for haematology analyzers [7], by analyzing pathological samples with high RBC or WBC concentrations followed by physiological samples. A patient specimen with a high RBC count was analyzed three times on the iQ200 analyzer, followed by another specimen with low RBC concentration for three times. Similarly, carry-over was evaluated by using three urine samples with WBC in urine and other three samples with physiological WBC counts. The carry-over was also estimated by using the iQ Positive Control (with fixed RBC, 1100 × E6/l) and Negative Control solution.

Between-run (day-to-day) and within-run (within-day) repeatabilities were determined by creating a *High Positive pool* from Positive Control, Focus and Calibrator solution, and a *Low Positive pool* by diluting the High Positive pool with iQ Diluent, and analyzing them 10 times each day for 10 working days (two weeks). The samples were stored refrigerated between the measurements. Additionally, within-run repeatability was obtained by analyzing natural urine particles in patient samples in duplicate for ten times. These results were calculated according to the NCCLS recommendation [8].

2.5. Statistics

Microsoft[R] Excel spreadsheets (Microsoft Corporation) were used for numerical data collection, cross-tabulations, graphs and regression analysis. Sensitivities, specificities, and Cohen's kappa coefficients [9] were also calculated by means of MS Excel spreadsheets. A *p*-value of 0.050 was used as the limit of statistical significance, equalling a 95% binomial confidence interval of proportions. We also assumed that Cohen's simple kappa coefficient is optimal if larger than 0.8, weighted kappa coefficient if larger than 0.85.

3. Results

3.1. Linearity

Linearity of counting was determined in the specified linearity range of the iQ200 analyzer (0–1000 cells × E6/l), by diluting serially the iQ Positive Control (1150 cells × E6/l) down to 12.5 cells × E6/l. Linearity was also tested by diluting serially two patient samples with high RBC counts (obtained ranges 27.5–1635 × E6/l and 13.5–647 × E6/l, respectively) and one sample with a high leukocyte count (created range 19–863 × E6/l). Linearity was shown at a squared coefficient of correlation (R^2) of 0.9874–0.9987 ($p \ll 0.001$) (Table 1).

3.2. Carry-over

The carry-over test was run with three different samples with hematuria (36, 420 and 4600 RBC × E6/l), and with three other samples with pyuria (27, 37 and 101 WBC × E6/l). After these positive specimens, none of the following physiological specimens in the series showed higher than physiological particle concentrations (all below 10 cells × E6/l). The serial test with the iQ Positive and Negative Control gave a similar result. Thus, no carry-over was observed.

3.3. Precision

Both within and between-run repeatability could be estimated by counting created control pools with high and low particle (fixed RBC) concentrations repeatedly (Table 2). In the assessment of results, the theoretical imprecision of particle counting, due to the Poisson distribution of events sampling must be considered as the analyzer samples a total volume of 2 µl. Theoretical imprecision is included in the table together with an imprecision ratio (observed/theoretical imprecision). Within-day repeatability exceeded the theoretical Poisson imprecision by 1.56 to 1.68-fold, while the between-day repeatability exceeded the theoretical Poisson imprecision by 2.25 to 2.70–fold. The ratio remained similar both at high and at low particle concentrations. To assess the precision in counting of different kinds of urine particles, within-day repeatability was estimated by repeated duplicate measurements of patient specimens. A ratio of 1.18–3.08 between observed and theoretical imprecision was obtained for most common urine particles, with observed CV's ranging from 4.0% (at WBC count 464.6 × E6/l) to 48.4% (at SQEP cell count 6.35 × E6/l), mostly depending on the particle concentration (Table 3). No difference in imprecision between the reported particle types was detected. The lower limit of quantification with iQ200 was about 20 – 30 particles × E6/l when using a criterion of between-day repeatability CV < 30%.

Table 1
Linearity of counting with Iris iQ200 Automated Urine Analyzer

Specimen	Number of serial dilutions	Regression line[a]	Coefficient (R^2)
iQ200 positive control	6	$y = 11.6x + 2.3$	0.9987
Patient urine 1 with RBC	7	$y = 26.0x - 28.3$	0.9983
Patient urine 2 with RBC	6	$y = 20.2x - 24.4$	0.9874
Patient urine 3 with WBC	6	$y = 27.8x - 0.8$	0.9936

[a] x = calculated, y = measured.

Table 2
Within-day and between-day repeatability obtained with control particle pools for iQ200 Automated Urine Analyzer

	High positive pool, mean = 761 particles × E6/l CV (%)	Low positive pool, mean = 21 particles × E6/l CV (%)
Day 1	5.24	28.3
Day 2	4.62	16.1
Day 3	4.63	23.7
Day 4	4.07	27.7
Day 5	5.98	21.5
Day 6	5.02	26.1
Day 7	3.06	22.2
Day 8	1.60	17.2
Day 9	5.52	40.2
Day 10	3.29	16.1
Within-day CV (%), mean of days 1–10	4.30	23.9
Between-day CV (%)	6.93	34.4
Theoretical imprecision (%)[a]	2.56	15.3
Imprecision ratio[b], within-day	1.68	1.56
Imprecision ratio[b], between-day	2.70	2.25

Counting was performed 10 times daily for 10 days.
[a] Calculated from Poisson distribution (since the analyzer counts a total volume of 2 μl, T = total count = 2 × mean count): $SD(T) = \sqrt{(T)} = \sqrt{(2 \times \text{mean count})}$; $CV(T) = SD(T)/T$.
[b] Ratio between observed and theoretical imprecision.

3.4. Method comparison against phase contrast microscopy

Comparisons were performed in two steps, first by APR alone, followed by an independent re-classification of captured images by the experienced users at Iris Diagnostics Inc. (E.C.). A total of 165–167 specimens could be compared both by phase contrast microscopy and Iris iQ200 APR alone, but due to technical file errors, a total of 163 samples were recovered for proper re-classification with iQ200 software by the user (E.C.). RBC, WBC, SQEP cells, Casts, and NSE cells were included in the quantitative comparisons (Figs. 1–5). The original APR results (series *A* in Figs. 1–5) and the results after the user intervention are shown (series *B* in Figs. 1–5), to show both the performance of the automated neural network algorithm alone, and improvements after human re-classification. In the graphs, theoretical imprecision of low particle counts had an effect on agreement at concentrations less than 20 particles × E6/l. Red blood cells (RBC) counting was essentially quantitative, but somewhat insensitive with a slope of 0.622 as compared to the reference phase contrast microscopy with automated algorithm alone ($r = 0.894$, $R^2 = 0.799$, $p \ll 0.001$; Fig. 1A). This could be improved to $r = 0.948$, ($R^2 = 0.898$, $p \ll 0.001$) by means of re-classification by a trained technologist (Fig. 1B). White blood cells (WBC) counting with APR alone showed a subpopulation of falsely high iQ200 counts, yielding a correlation coefficient $r = 0.885$, ($R^2 = 0.783$, $p \ll 0.001$; Fig. 2A). After re-classification by a trained technologist, the WBC counting improved to $r = 0.978$ ($R^2 = 0.956$, $p < 0.001$) for WBC counting (Fig. 2B). The third best, still essentially quantitative counting result was obtained with squamous epithelial cells (SQEP) resulting in $r = 0.904$ ($R^2 = 0.818$, $p \ll 0.001$) for the APR algorithm alone

Table 3
Within-run repeatability in counting different urine particles

Particle type[a]	Mean count (× E6/l)	Observed CV (%)[b]	Theoretical CV (%)	Imprecision ratio[c]
RBC 1	12.25	32.6	20.2	1.61
RBC 2	1246	6.2	2.0	3.08
WBC 1	114.9	10.4	6.6	1.58
WBC 2	464.6	4.0	3.3	1.23
SQEP	6.35	48.4	28.1	1.72
NSE	6.65	32.4	27.4	1.18
Bacteria	45.9	30.0	10.4	2.88
Yeast	32.7	25.1	12.4	2.03

[a] The abbreviations are: RBC, red blood cells (patients 1 and 2); WBC, white blood cells (patients 1 and 2); SQEP, squamous epithelial cells; NSE, non-squamous epithelial cells.
[b] Mean within-run repeatability calculated from duplicate ten measurements during the same day.
[c] Ratio = observed / theoretical Poisson imprecision.

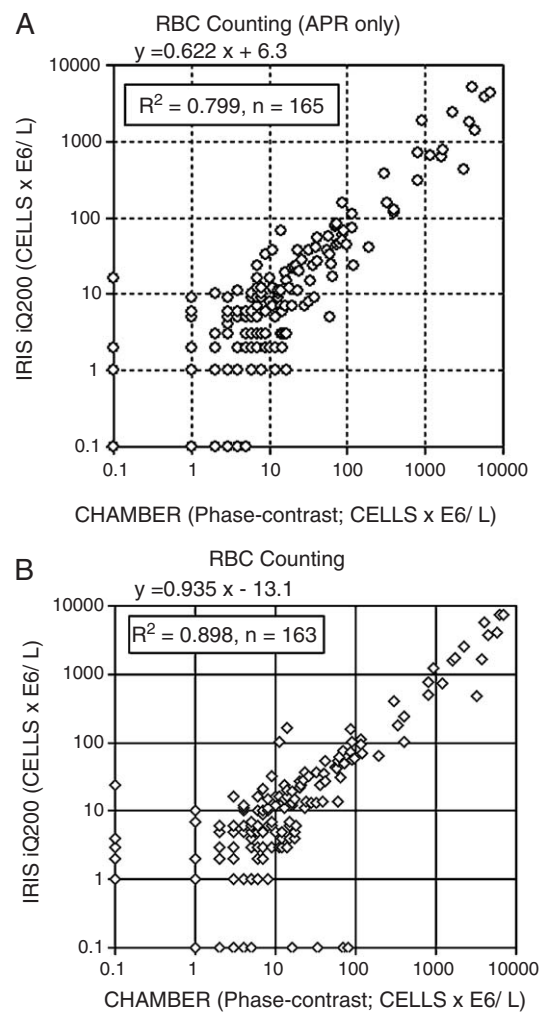

Fig. 1. Comparison of RBC counting by Iris iQ200 against visual chamber with phase contrast optics. (A) Analysis with Automated Particle Recognition (APR) alone, (B) after re-classification by human user.

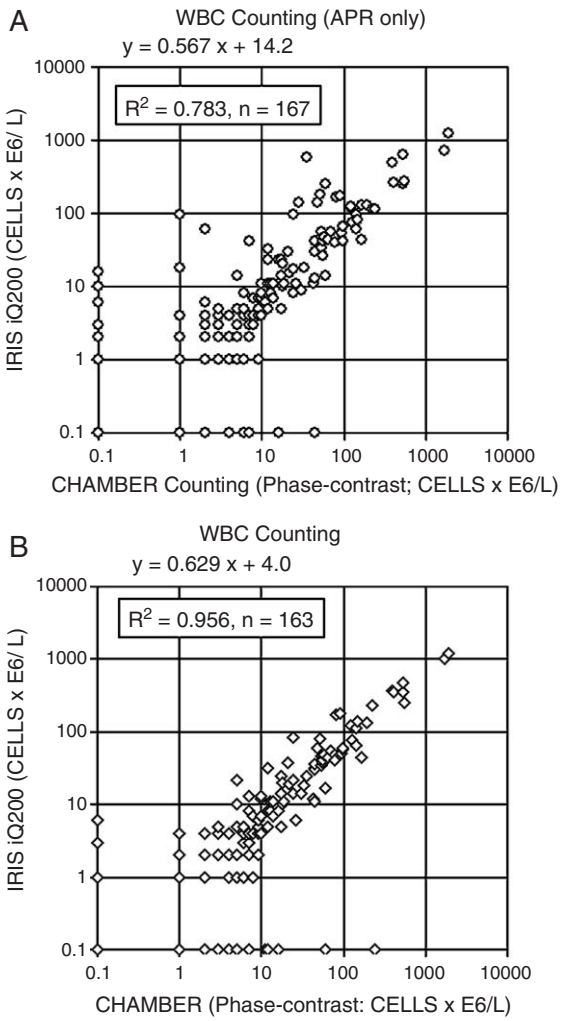

Fig. 2. Comparison of WBC counting by Iris iQ200 against visual chamber with phase contrast optics. (A) Analysis with Automated Particle Recognition (APR) alone, (B) after re-classification by human user.

(Fig. 3A). The average count becomes even more quantitative after user reclassification ($y=1.002\times+0.14$; Fig. 3B), but the correlation ($r=0.927$, $R^2=0.860$, $p\ll0.001$) did not improve essentially because counts of epithelial cells are low in urine in general, the highest values hardly reaching 100 × E6/L.

Renal elements are clinically significant findings in urine. Cast counting was evaluated by combining both hyaline (mostly seen) and non-hyaline casts together because of rarity of non-hyaline casts in our material. Thus, the term "cast" included both hyaline and non-hyaline casts in this study. Detection of casts failed with APR alone ($R^2=0.000$, $p=0.986$; Fig. 4A), but was essentially improved in the reclassification by the user ($r=0.856$, $R^2=0.732$; $p\ll0.001$; Fig. 4B). Our study included only 3 specimens having more than 10 casts × E6/L in phase contrast microscopy. The regression analysis may not provide a good performance evaluation for casts with this limited collection of specimens. Small epithelial cells (both renal tubular and transitional cells) were classified into one group of NSE cells. Low counts of these cells observed with phase contrast microscopy were partially detected by the APR classification ($r=0.585$, $R^2=0.342$; $p\ll0.001$; Fig. 5A) and somewhat better after re-classification by a trained technologist ($r=0.706$, $R^2=0.499$; $p\ll0.001$; Fig. 5B), but not to an optimal level. The regression equation suggests that 3 out of 10 NSE cells were detected in the average (slope = 0.301). The iQ200 reported a higher number of negative specimens for NSE cells than reported by phase contrast microscopy. Again, our study included only 5 specimens having more than 10 × E6/l NSE cells. The regression analysis may not provide a good performance evaluation for NSE with this limited collection of specimens. The rare oval fat bodies (degenerated renal tubular cells) were not reported separately.

For comparing bacteria, yeasts and crystals, we used an ordinal scale result in phase contrast microscopy (negative, 1+, 2+ or 3+) and a quantitative count obtained by iQ200 analyzer. Two particles × E6/l were considered as a limit of positivity for bacteria and yeast, because of the clinical need to detect these microbes sensitively. A cut-off of 10 particles × E6/l was used for crystals to allow statistical uncertainty. Ordinal scale data were organized in a tabular form, including calculations for sensitivity, specificity and Cohen's kappa coefficients (Table 4A

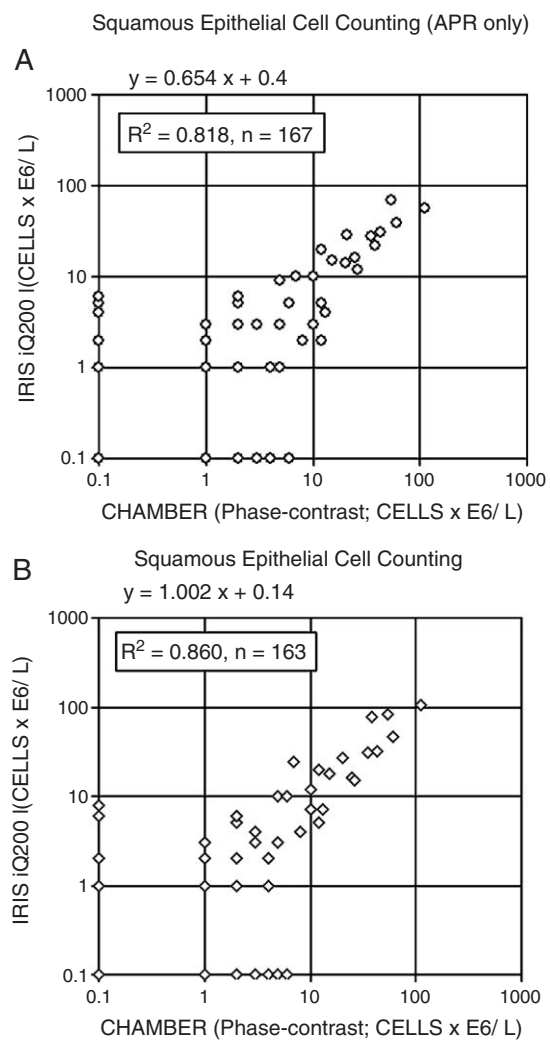

Fig. 3. Comparison of squamous epithelial cell counting by Iris iQ200 against visual chamber with phase contrast optics. (A) Analysis with Automated Particle Recognition (APR) alone, (B) after re-classification by human user.

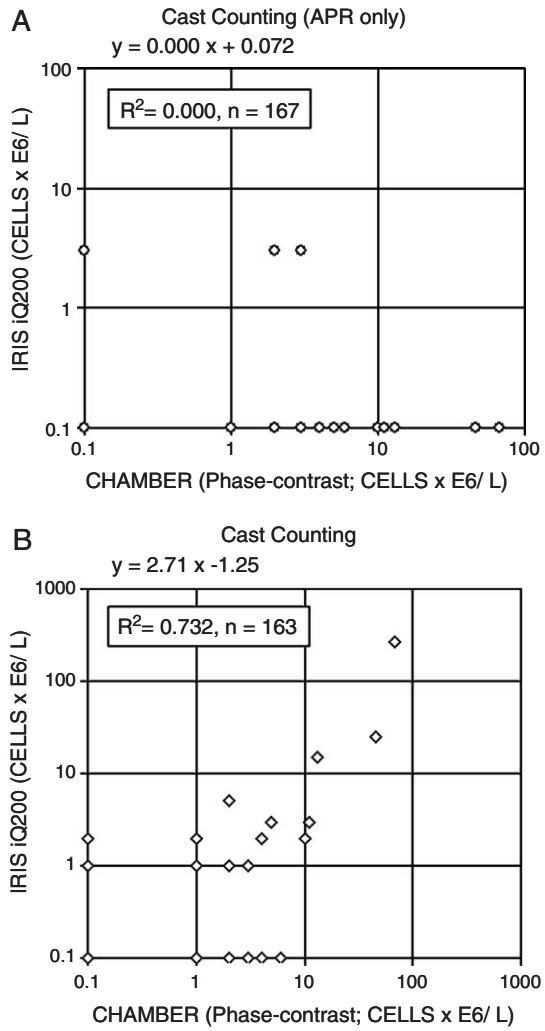

Fig. 4. Comparison of cast counting by Iris iQ200 against visual chamber with phase contrast optics. (A) Analysis with Automated Particle Recognition (APR) alone, (B) after re-classification by human user.

with by APR alone, Table 4B after user-reclassification). In two cases, no result was obtained for yeasts by the iQ200 analysis. A sensitivity of 42.9% was reached in the detection of bacteria by APR alone with a specificity of 96.4% against visual phase contrast microscopy (Table 4A). After re-classification by a trained technologist, sensitivity improved to 64.3%, detecting all specimens with high bacterial counts, while specificity remained at 97.0% (Table 4B). Very few specimens with yeast were observed at HUSLAB. Out of the 165 specimens that could be assessed for yeast both by phase contrast microscopy and iQ200, only 6 true positive cases were found. The APR system alarmed 58 times for yeast at the cut-off of 2 particles × E6/l, including 4/6 true positive cases. The amount of false positives remained fairly high even at the cut-off level of 10 particles × E6/l (Table 4A). The manufacturer indicated that the iQ200 could be programmed with a higher threshold to reduce false alarms for yeast. After re-classification by a trained technologist specificity improves to 96.8% while sensitivity remained at 66.7% (Table 4B). Only 4 specimens with crystals were seen in phase contrast microscopy, out of which only two remained for the careful user re-evaluation of 163 specimens. The other two could not be recovered because of a specimen identification error (Table 4B). After confirmation by a trained technologist, we obtained a specificity of 93.2% and a sensitivity of 50%, resulting in a low positive predictive value for crystal reports by iQ200 in our patient material. This was exaggerated by the small amount of true positive cases.

Automated and user-edited results from iQ200 analysis were compiled to show the sensitivity and specificity at a given cut-off limit against the reference phase contrast microscopy for each urine particle type. In addition, routine results obtained by bright field microscopy of uncentrifuged specimens by several laboratory technologists at HUSLAB were compared to the reference microscopy (Table 5). Due to inherent statistical imprecision, 100% sensitivity with 100% specificity cannot be reached at low particle counts. The arbitrary upper reference limits were, however, considered informative in the assessment of positivity. In counting RBC, the iQ200 analysis after human reclassification was slightly more sensitive, 80.2% [73.9–

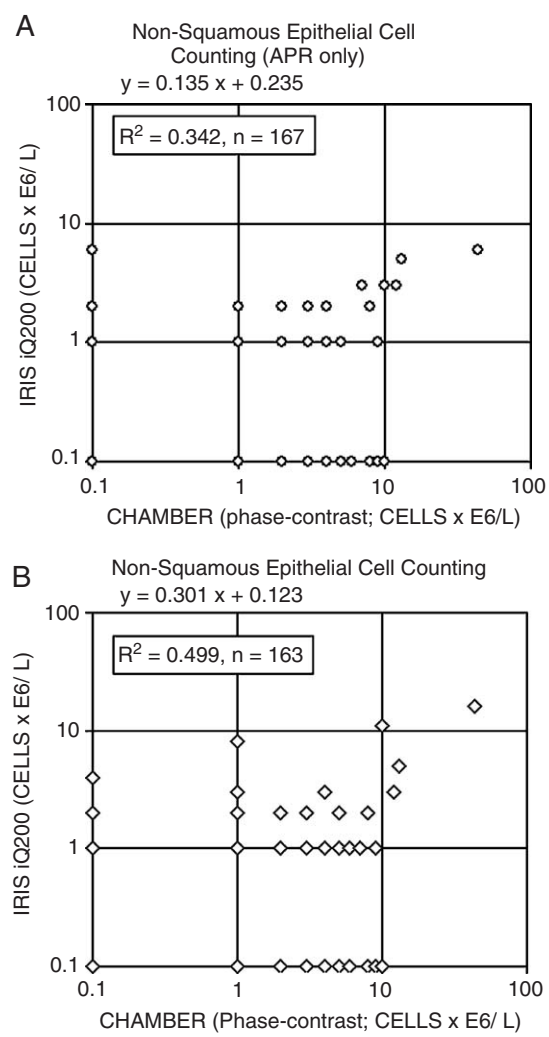

Fig. 5. Comparison of non-squamous (small) epithelial cell counting by Iris iQ200 against visual chamber with phase contrast optics. (A) Analysis with Automated Particle Recognition (APR) alone, (B) after re-classification by human user.

Table 4A
Comparison of iQ200 and reference phase contrast microscopy, ordinal scale results by automated particle recognition (APR) alone

Bacteria	Chamber (ordinal scale)				
iQ200 (×E6/l)	Negative	1+	2+	3+	Total
0–1	134	8	3	5	150
2–9	4	3		4	11
10–49	1			4	5
50–more				1	1
Grand total	139	11	3	14	167
Sensitivity =	42.9%				
Specificity =	96.4%				
Kappa (weighted) =	0.459				

Yeast	Chamber (ordinal scale)			
iQ200 (×E6/l)	Negative	1+	2+	Total
0–1	105	1	1	107
2–9	29	1		30
10–more	25	2	1	28
Grand total	159	4	2	165
Sensitivity =	66.7%			
Specificity =	66.0%			
Kappa (simple) =	0.055			

Crystals	Chamber (ordinal scale)		
iQ200 (×E6/l)	Negative	Positive	Total
0–9	145		145
10–more	18	4	22
Grand total	163	4	167
Sensitivity =	100.0%		
Specificity =	89.0%		
Kappa (simple) =	0.278		

86.5%, binomial confidence interval=CI 95%] than routine bright field microscopy 65.9% [58.5–73.3%, CI 95%], while APR classification alone did not differ significantly from either of them with a sensitivity of 73.2% [66.3–80.1%, CI 95%]. Sensitivity of RBC detection by iQ200 could not be improved by including specimens of low specific gravity with a test strip result positive for RBC additionally (assuming that lysed RBC could be detected by pseudoperoxidase activity; data not shown), indicating that the reduced sensitivity was related to technical capabilities of the iQ200 instrument and partially to statistical imprecision of low counts. Specificity of RBC counting was highest with routine bright field microscopy 100% [95.4–100%, CI 95%], but lower with APR alone 88.1% [83.1–93.1%, CI 95%] or after user re-classification 82.9% [77.0–88.8%, CI 95%]. WBC counting performed equally well against reference phase contrast microscopy by all procedures, also with the APR alone. Detection of SQEP cells was slightly better by both iQ200 procedures than by routine bright field microscopy, but the sensitivities did not reach statistical significance. It is noteworthy that detection of (combined) CASTS was difficult by routine bright field microscopy, showing a sensitivity of 26.3% [19.5–33.1%, CI 95%] only. Automated recognition of casts by iQ200 analyzer had a sensitivity of 10.5% [5.7–15.3%, CI 95%], but improved to 42.1% [34.3–49.9%, CI 95%] by human reclassification.

In the detection of NSE cells, a low sensitivity of 10.4% [5.7–15.1%, CI 95%] was seen in bright field microscopy, which increased by automated iQ200 analysis and reclassification, reaching a sensitivity of 25.5% [18.7–32.3%, CI 95%]. Thus, renal elements were detected sensitively enough by neither routine bright field microscopy nor by the instrument, as compared to the reference phase contrast procedure. In the detection of bacteria, the iQ200 analyzer with APR alone reached a sensitivity of 42.9% [35.2–50.6%; CI 95%], but after user reclassification a sensitivity of 64.3% [56.8–71.8%, CI 95%] as compared to 71.4% [64.4–78.4%, CI 95%] in routine bright field microscopy. Detection of yeasts was comparable to that with bright field microscopy with our low number of positive cases. Classification of crystals suffered from false positive cases in iQ200 analysis, not detected in visual microscopic techniques.

3.5. Feasibility

To simplify the study, the evaluators were not trained on editing (re-classification) of particles with the iQ200 software, as the manufacturer recommends it to the users of the iQ200. The automated part of the analyzer was easy to learn. Daily maintenance included a short service action once a day. The system needed calibration only once a month and the daily routine service operation including a run with wash and diluent

Table 4B
Comparison of iQ200 and reference phase contrast microscopy, ordinal scale results after editing by an experienced operator

Bacteria	Chamber (ordinal scale)				
iQ200 (×E6/l)	Negative	1+	2+	3+	Total
0–1	131	8	2		141
2–9	2	3	1	1	7
10–49	1				1
50–more	1			13	14
Grand total	135	11	3	14	163
Sensitivity =	64.3%				
Specificity =	97.0%				
Kappa (weighted) =	0.855				

Yeast	Chamber (ordinal scale)			
iQ200 (×E6/l)	Negative	1+	2+	Total
0–1	152	2		154
2–9	2	2		4
10–more	3	1	1	5
Grand total	157	5	1	163
Sensitivity =	66.7%			
Specificity =	96.8%			
Kappa (simple) =	0.449			

Crystals	Chamber (ordinal scale)		
iQ200 (×E6/l)	Negative	Positive	Total
0–9	150	1	151
10–more	11	1	12
Grand total	161	2	163
Sensitivity =	50.0%		
Specificity =	93.2%		
Kappa (simple) =	0.124		

Table 5
Summary of the diagnostic performance of iQ200 and routine bright field microscopy against the reference phase contrast microscopy

Particle type	Limits for positivity	Number of positive cases	Routine visual microscopy with bright field optics[a] (n=167)		Automated particle classification alone[a] (n=167)		Re-classified iQ200 results by operator[a] (n=163)	
			Sensitivity	Specificity	Sensitivity	Specificity	Sensitivity	Specificity
RBC	10	81	65.9% (a)	100% (b,c)	73.2%	88.1% (b)	80.2% (a)	82.9% (c)
WBC	10	71	82.2%	93.6%	83.6%	92.6%	80.3%	96.7%
SQEP	10	16	70.6%	100%	76.5%	99.3%	75.0%	98.0%
CASTS	2	19	26.3% (a,b)	99.3%	10.5% (a,c)	98.6%	42.1% (b,c)	98.6%
NSE	2	51	10.4% (a)	95.0%	18.8%	96.6%	25.5% (a)	94.6%
Bacteria	2/ordinal[b]	28	71.4% (a)	97.1%	42.9% (a,b)	96.4%	64.3% (b)	97.0%
Yeast	2/ordinal[b]	6	66.7%	99.4% (a)	66.7%	66.0% (a,b)	66.7%	96.8% (b)
Crystals	10/ordinal[b]	2	50.0% (a)	100% (c)	100% (a,b)	89.0% (c)	50.0% (b)	93.2%

[a] Statistical significance of difference in sensitivities or specificities against the phase contrast reference microscopy was compared pairwise between Routine bright field microscopy, Automated particle classification alone, and Re-classified results of iQ200 instrument; significantly different pairs are indicated with alphabets on each line. Significant difference = non-overlapping 95% binomial confidence interval (CI) of proportion (sensitivity or specificity) between the two procedures (equals to $p<0.05$).
[b] Ordinal scale was used in visual microscopic results.

solutions, focusing, and running controls took only about 10 min. The samples could be run with or without barcodes on the analyzer. Time for getting one result from the analyzer was about 1 min. If a long series of samples was run the analysis proceeded normally but the results appeared on the screen more slowly. The analyzer was quite silent and did not produce heat.

Noticed disadvantages included the following: iQ200 captures particle images from unstained urine. Some particles may be difficult to identify if laboratory technologists are used seeing the particles after staining in (either bright field or phase contrast) visual microscopy. Although squamous epithelial cells are fairly sharp, small particles may appear out-of-focus and operators cannot focus them better as it is possible in a visual microscope. The iQ200 software we evaluated did not display the actual size of an image; the software available at the time of this writing displays a measuring bar together with a shown image. Increasing ten-fold the total volume counted (from 2 to 20 μl) could overcome the inherent Poisson imprecision of low particle counts in urine, i.e., as soon as the computing can deal with the increased information flow. Ideally, reference microscopy should also be performed with high enough particle counts if practically feasible [5].

Adaptation to small sample volumes is important. The iQ200 system could use the 10-ml tubes used in Finland. An approach to paediatric samples is a dilution that can be made with the Iris Diluent solution. The system could automatically calculate the dilution factor if a barcode label indicating the dilution ratio was placed on the tube. Dilutions are user-definable.

4. Discussion

Routine urine particle analysis stays historically behind haematology in the development of automation of emergency laboratory services, because patient urine is a more difficult matrix than blood, clinically significant particles vary 100-fold in their size (from about 1 to 100 μm in diameter), and because urine particle concentrations are generally low. Infectious diseases of kidneys and urinary tract are associated with pyuria and bacteriuria (occasionally with yeast). Non-infectious diseases often present themselves by hematuria. In addition, kidney-derived particles (casts, renal tubular cells) reflect the presence of renal disease together with proteinuria (excretion of albumin and tubular proteins). Automated particle counting carries a promise of improved detection and quantitative monitoring of diseases of the urinary tract, because visual microscopy is prone to human and technical errors [10], as seen in our routine bright field results as well (Table 5). In addition, improving efficiency required from basic laboratory services and shortage of skillful laboratory professionals favours development of automation.

European urinalysis guidelines divided reporting of urine particles into basic and advanced levels depending on the needed detail of differentiation [11]. The basic level included differentiation into RBC, WBC, squamous and non-squamous ("small") epithelial cells, hyaline and non-hyaline casts, bacteria and yeast, *Trichomonas*, spermatozoa, lipids and crystals. The basic particles that should be counted most accurately include RBC, WBC, and (uropathogenic) bacteria. Squamous epithelial cells indicate contamination that affects interpretation of bacterial culture. Renal elements are useful in the follow-up of kidney diseases, but do not replace sensitive proteinuria and hematuria measurements at the screening stage [12].

Counting of RBC by Iris iQ200 analyzer succeeded at a squared coefficient $R^2=0.799$ ($r=0.894$) with the APR alone, and at a coefficient $R^2=0.898$ ($r=0.948$) after re-classification by trained technologist (Fig. 1) as is recommended by the manufacturer. Similarly quantitative correlations were obtained for WBC with a squared coefficient $R^2=0.783$ ($r=0.885$) with APR alone, and at a coefficient $R^2=0.956$ ($r=0.978$) after re-classification by human editor (Fig. 2). These correlations show a quantitative recognition of most RBC and WBC in urine even automatically. In our study, sensitivity of RBC detection could not be improved by test strip results, perhaps due to short delays (less than 3 h) of our counting (data not shown).

Counting of SQEP cells was essentially quantitative as well, with a squared coefficient $R^2=0.818$ ($r=0.905$) with APR alone, and $R^2=0.860$ ($r=0.927$) after user re-evaluation, despite

the fact that most specimens had SQEP cells at concentrations less than 10 × E6/l (Fig. 3). This result was better than $r = 0.722$ we reached earlier with the flow cytometer UF-100 [3].

Other evaluations on the performance of iQ200 instrument have already been published. Lamchiagdhase et al. studied carefully urine specimens from 280 outpatients, comparing iQ200 to visual microscopy of urine sediments counted in under a coverslip according to NCCLS protocol, or in a standardized commercial chamber [18]. Correlations of iQ200 results after technologist's review to those of chamber counting of WBC, RBC and SQEP were 0.816, 0.976 and 0.877, respectively. No significant differences were found between the levels obtained from iQ200 and reference chamber counts, in the opposite to routine microscopic counts that were markedly lower. The results are well in line with our study, despite that a centrifugation step reduced correlations of iQ200 to visual procedures. Sensitivities of iQ200 counting were 88.8% for WBC, 84.4% for RBC, and 86.3% for SQEP, respectively, using a cut-off of 27 cells × E6/l. Specificities were 93.6%, 93.1% and 98.4%, in the respective order. These figures correlate well with our results (Table 5), but the percentages are higher due to both higher cut-offs (we used 10 cells × E6/l for WBC and RBC), and possibly bright-field microscopy that lost some of the visual observations.

Wah et al. compared the iQ200 to direct bright field microscopy in Fuchs–Rosenthal chambers and uncentrifuged specimens [19]. Their correlations between the procedures for RBC, WBC and epithelial cells were very good; r values ranging from 0.940 to 0.959. The slope from their regression analysis was better for WBC (0.81) than that of our Finnish study even after re-classification (0.629; Fig. 2B), perhaps because we could detect degenerating WBC better than the instrument using phase contrast optics. The iQ200 instrument added some non-WBC into the counts (Fig. 2A) that could be removed by the human operator (Fig. 2B). A third comparison was made by Alves et al. between standardized urine sediment and iQ200 microscopy analyzer [20]. Because of centrifugation step and an ordinary scale reporting, correlations remained lower than those in our study [20].

The other automated microscopic technique, flow cytometric counting with the instrument UF-100 (Sysmex) succeeded in our hands at a squared coefficient $R^2 = 0.774$ ($r = 0.880$) for RBC and at a coefficient $R^2 = 0.964$ ($r = 0.982$) for WBC against visual chamber counting [3]. Fenili and Pirovano showed a correlation of $r = 0.833$ for RBC and $r = 0.934$ for WBC between UF-100 and sediment microscopy [13]. Sensitivity and specificity figures suffered from imprecision related to low particle counts below 20 particles × E6/l, but were decided to stay at the estimated upper reference limits of healthy individuals (Table 5). Wah et al. [19] do not report their limits of normal used in the sensitivity estimates of 76–86% with specificities exceeding 92% for RBC, WBC and SQEP cells.

In classification of RBC, small or dysmorphic RBC were discerned with some difficulty by the automated microscopy, but also in routine bright field visual microscopy. Some authors have recommended detection of haemoglobin from lysed RBC, and esterase from lysed WBC (granulocytes) using test strip chemistry as a supplement of particle counting [14,15]. The iQ200 analyzer can be connected to a test strip analyzer and the test strip results can be seen together with the iQ200 results. Test strip supplementation is important especially in case of delayed examination and non-preserved specimens.

Counting of casts and NSE cells succeeded at a moderate level by iQ200. After re-classification by a trained technologist, results were essentially improved but remained non-optimal, $R^2 = 0.732$ ($r = 0.856$) for (combined) casts (Fig. 4), and $R^2 = 0.499$ ($r = 0.706$) for NSE cells (Fig. 5), respectively. Difficulty in detecting renal elements was also reported by Lamchiagdhase et al. [18], although blaming mostly false negative results. Our phase contrast microscopy would indicate that the iQ200 instrument can see these particles better than bright field optics in routine microscopes, since we had a sensitivity problem, not a specificity problem (Table 5). We have studied earlier the influence of particle concentration in the performance of UF-100: sensitivity varied within a range of 18–69% as the cut-off ranged from 1 to 10 renal particles × E6/l; [3]. It is to be noted that the routine bright field microscopy at HUSLAB was even less sensitive in detecting these elements, since phase contrast optics are needed to detect all hyaline casts, and careful differentiation is needed to discriminate small epithelial cells from leukocytes (Table 5). The CCD camera of iQ200 is capable of capturing cast images better than a usual bright field microscopist, but the APR has a difficulty in classifying casts. The iQ200 instrument has, however, the option of visual re-classification of captured images. This results in better performance than obtained with the average routine bright field microscopy or a current flow cytometer. Targeted, trained visual microscopy with phase contrast optics (and supravital staining to facilitate differentiation of cells) is recommended for those patients for whom the most sensitive detection of renal elements is important. In our experience, this is requested for 1–10% of routine specimens at tertiary level general hospitals in Finland.

Precision was tested by using both fixed RBC (control pools; Table 2) and different particles in patient specimens (within-day repeatability only; Table 3). The inherent imprecision related to Poisson distribution of particles in solutions was "subtracted" by dividing the obtained CV's with the theoretical Poisson CV. The obtained imprecision ratio remained 2 to 3-fold against the theoretical value, actual observed CV's ranging from 2–10% at high concentrations to 15–50% at low particle concentrations for all particle types tested (Tables 2 and 3). If a maximum allowable between-day precision of 30% is used as a criterion, the lower limit of detection of iQ200 is about 20–30 particles × E6/l. These figures are essentially similar to those reported by Wah et al. [19]. The manufacturer indicates that the iQ200 APR software is not trained to recognize fixed red blood cells contained in the iQ Positive control. Thus the between-day and within-day repeatability values quoted above include both recognition repeatability and counting precision for fixed RBC. The repeatabilities related to different natural particles are shown in Table 3, but only a within-day study was possible with non-preserved material. Obtained figures theoretically depend on the particle concentrations and total volumes counted. For UF-100 flow cytometer, the following between-day repeatabilities have been

reported: for RBC 4.1% at 195 × E6/l, and for WBC 2.8% at 195 × E6/l [16]. Note, however, that these mean concentrations are at the high pathological range.

Clinical ranges of urine particle concentrations vary 10 to 1000-fold from healthy to diseased states. As a minimum goal, a 10-fold difference between particle concentrations of two specimens should be recognized in patient care. An allowable total error of each count would then be $10/(1.96\times\sqrt{2})\approx 3.6$ – fold, by applying the formula of critical difference (at $p<0.050$) [17]. The observed analytical accuracy was maximally 0.5-fold (50%) at the low concentration range, assuming that the particles were reported correctly. Because of improved standardization, automated procedures now allow improved estimates of quality specifications for both analytical and pre-analytical processes.

Detection of bacteria had a sensitivity of 42.9%, and a specificity of 96.4%, with APR alone (Table 4A), but improved to a sensitivity of 64.3% with a specificity of 97.0% against reference phase contrast microscopy after re-classification by human user (Table 4B).

Both Lamchiagdhase et al. [18] and Alves et al. [20] have reported that presence of bacteria, and budding yeast should be confirmed under the microscope.

The sensitivity of UF-100 (using the earlier software with separate BACT and H-BACT channels) has been 51% with a specificity of 97% against phase contrast microscopy in chamber (T.K., unpublished results). No rapid technique, not even phase contrast microscopy is sensitive enough as compared to bacterial culture if the low-count bacteriuria is to be detected [3]. In the present evaluation, several clearly uropathogenic species and significant colony counts in the culture remained undetected by particle counting (data not shown). The users of iQ200 have to try to improve their detection of bacteria to maximize rapid diagnostics whenever possible, since all rapid specific reports on bacteria help in initiating anti-microbial treatment. Finally, a large proportion of bacteria still remain to be shown by the culture alone.

The iQ200 falsely reported yeast or crystals too often in our evaluation (Table 4A). These particles are not key elements of basic particle analysis, and could be eliminated by adjusting the corresponding thresholds in routine iQ200 reports. Yeast is usually diagnosed in routine bacterial culture. True pathological crystals are rare and can be diagnosed after a specific request if needed. Innocent crystallization of urates or phosphates often follows refrigeration of urine specimens.

Image storage and later re-evaluation, even consultation or teaching, are important features of an automated microscope that makes the transition from “wet urinalysis” to “dry urinalysis” at bench top possible. Technical development of CCD cameras and image processing will probably bring these images gradually closer to visual phase contrast optics to allow maximal detection of all formed elements in urine.

5. Conclusions

The iQ200 system was reliable in counting RBC, WBC, and squamous epithelial (SQEP) cells. It also identified a fraction of bacteria and renal elements (casts and NSE cells) with the assistance of a trained technologist. The automated analysis of uncentrifuged urine with iQ200 eliminates manual sample preparation, standardizes the procedure of counting, and provides storage and review of captured images. The instrument does not replace a properly trained technologist but rather helps her/him to be more efficient in reporting urine particle findings. Technical improvement is needed to improve the specificity of yeast and crystal detection. Image original size, contrast and sharpness were occasionally less favourable than desired. In most cases, visual microscopy seems not to be needed for confirmation of findings. Instrument size and walk-away capabilities have been markedly improved from the preceding 900UDX series of Iris instruments.

Acknowledgments

Iris Diagnostics Inc. provided financial and technical assistance for this evaluation, in addition to an encouragement of true scientific evaluation of the performance data.

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